A prospective study of dietary acrylamide intake and the risk of endometrial, ovarian, and breast cancer.

BACKGROUND: Acrylamide, a probable human carcinogen, was detected in various heat-treated carbohydrate-rich foods in 2002. The few epidemiologic studies done thus far have not shown a relationship with cancer. Our aim was to investigate the association between acrylamide intake and endometrial, ovarian, and breast cancer risk. METHODS: The Netherlands Cohort Study on diet and cancer includes 62,573 women, aged 55-69 years. At baseline (1986), a random subcohort of 2,589 women was selected using a case cohort analysis approach for analysis. The acrylamide intake of subcohort members and cases was assessed with a food frequency questionnaire and was based on chemical analysis of all relevant Dutch foods. Subgroup analyses were done for never-smokers to eliminate the influence of smoking; an important source of acrylamide. RESULTS: After 11.3 years of follow-up, 327, 300, and 1,835 cases of endometrial, ovarian, and breast cancer, respectively, were documented. Compared with the lowest quintile of acrylamide intake (mean intake, 8.9 mug/day), multivariable-adjusted hazard rate ratios (HR) for endometrial, ovarian, and breast cancer in the highest quintile (mean intake, 40.2 mug/day) were 1.29 [95% confidence interval (95% CI), 0.81-2.07; P(trend)=0.18], 1.78 (95% CI, 1.10-2.88; P(trend)=0.02), and 0.93 (95% CI, 0.73-1.19; P(trend)=0.79), respectively. For never-smokers, the corresponding HRs were 1.99 (95% CI, 1.12-3.52; P(trend)=0.03), 2.22 (95% CI, 1.20-4.08; P(trend)=0.01), and 1.10 (95% CI, 0.80-1.52; P(trend)=0.55). CONCLUSIONS: We observed increased risks of postmenopausal endometrial and ovarian cancer with increasing dietary acrylamide intake, particularly among never-smokers. Risk of breast cancer was not associated with acrylamide intake.